

A Unified Kinematic Representation Enables Reusable Biological Joint Moment Estimation

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Abstract—Objective: Data-driven models that estimate physiological states, particularly biological joint moments, are widely used in exoskeleton control. However, these estimators are often coupled to device-specific sensor configurations, limiting controller transfer and the use of open-source biomechanics datasets. **Methods:** We proposed joint kinematics as an intermediate representation that decouples hardware-specific sensing from downstream biological joint-moment estimation. A joint-moment estimator using joint angles and angular velocities was trained exclusively on open-source biomechanics data and evaluated using kinematics from a hip exoskeleton, knee exoskeleton, and inertial measurement unit (IMU) sensor suite during level-ground, ramp-ascent, and ramp-descent walking. **Results:** The estimator achieved an RMSE of 0.169 Nm/kg and R^2 of 0.801 using hip exoskeleton kinematics, 0.192 Nm/kg and 0.615 using knee exoskeleton kinematics, and 0.147 Nm/kg and 0.852 using kinematics derived from IMUs across bilateral hip, knee, and ankle joints. **Conclusion:** Joint kinematics enabled an estimator trained only on open-source data to operate across distinct wearable platforms. **Significance:** This framework may reduce target-device data collection and support transferable biological joint-moment estimation for exoskeleton control and wearable biomechanics.

Index Terms—Biological joint moment estimation, joint kinematics, device-agnostic exoskeleton control

I. INTRODUCTION

Lower-limb exoskeletons and other wearable robotic systems can augment movements [1], [2], restore mobility [3]–[5], and support independence for individuals with impaired locomotion [6]–[8]. Because movement patterns and mechanical demands vary across users, speeds, terrains, and tasks, effective assistance generally requires adaptive control rather than a single fixed strategy [2], [4], [9], [10]. Such controllers adjust assistance based on estimates of the user’s biomechanical state, including joint kinematics, contacts, gait phase, and joint moments.

Many exoskeleton controllers represent this state using discrete variables such as gait phase or terrain class [11]–[13]. Although useful for dividing locomotion into phases or tasks, these representations do not fully capture continuous changes in mechanical demand. Biological joint moments instead quantify the net mechanical contribution generated by the user’s musculoskeletal system at each joint, providing a continuous and mechanically interpretable measure of locomotor demand. Data-driven joint moment estimates have consequently been

used for subject-independent and task-agnostic exoskeleton control across diverse activities [9], [14], [15].

Inverse dynamics has been the standard to obtain biological joint moments, but this method requires motion capture and forceplate systems, only available in high-end gait laboratories, motivating the use of wearables and machine learning as an alternative combination for estimation of these quantities. Existing data-driven estimators commonly operate directly on device-specific signals, including inertial measurement units (IMUs), motor encoders, and pressure sensors [9], [16]–[20]. These approaches can achieve accurate estimates when deployment and training use the same sensing configuration [9], [14], [15]. However, they are generally overfitted to specific systems because the learned models also capture differences in sensor modality, placement, orientation, calibration, attachment, and preprocessing. This can alter the input distribution, limiting transfer to devices that do not reproduce the original inputs.

Prior work has addressed related this transfer challenge through online model adaptation and sensor-signal transformation. Kang et al. [4] adapted a gait-phase estimator to new users and locomotion conditions across exoskeleton platforms by transforming sensor signals between devices. Deep domain adaptation has similarly addressed sensor-domain mismatch by aligning representations between source and target wearable configurations [17]. Although these approaches can reduce or eliminate the need for labeled target-domain biomechanics, they still require recordings from the target user or sensing configuration. Changes in hardware, sensor placement, or preprocessing may therefore require new target-domain data and another adaptation procedure. This motivates a complementary strategy in which the downstream estimator operates on a common physical representation rather than hardware-specific signals.

Hardware-specific inputs also limit the use of open-source biomechanics data, in which, they provide extensive lower-limb kinematics, ground-reaction forces, and inverse-dynamics moments across participants and locomotor conditions [16], [21]–[29]. These resources can support generalizable estimators and biomechanical foundation models without requiring each group to reproduce the original laboratory data collection [30]. However, their wearable signals rely on the original sensing configurations and cannot represent every custom exoskeleton or future device. The significant limitation is the absence of a generalized input representation to reuse existing kinematics and inverse-dynamics labels across wearable systems.

Joint kinematics, such as joint angles and angular velocities,

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provide a potential intermediate representation as they can conveniently and accurately be obtained from motor encoders, segment-mounted IMUs, and inertial motion-capture systems [31]–[33]. In the proposed framework, each platform converts its available signals into a unified kinematic representation, which a downstream model maps to biological joint moments (Fig. 1). Because these inputs have consistent biomechanical meaning across platforms, the joint moment estimator becomes reusable across devices, in which the joint angles and angular velocities are available.

In this study, we investigated whether wearable-derived joint kinematics can serve as a common intermediate representation for biological joint moment estimation across heterogeneous sensing platforms. We trained causal temporal convolutional models exclusively on open-source kinematics and inverse-dynamics moments and then applied them, without target-device fine-tuning, to a bilateral hip exoskeleton, a unilateral knee exoskeleton, and an IMU sensor suite. We evaluated localized single-joint estimation, whole-limb multi-joint estimation, the influence of kinematic input quality, and the benefit of adjacent-joint context.

To the best of our knowledge, this study provides the first demonstration of a biological joint-moment estimator trained exclusively on open-source biomechanics data and deployed across previously unseen wearable platforms without target-device training, adaptation, or fine-tuning. We further validate this approach on physical hip and knee exoskeletons, demonstrating that a common kinematic representation can support joint-moment estimation across distinct hardware configurations. Despite using no data from the target devices during training, the resulting estimation performance is comparable to prior device-specific approaches trained directly on the hardware on which they were evaluated.

II. METHODS

A. Open-Source Data for Model Training

All biological joint moment estimators were trained using publicly available biomechanics data (Fig. 1(a)). The curated database combined level-ground, ramp-ascent, and ramp-descent walking trials from the datasets used in Camargo et al. [21], Scherpereel et al. [16], and Molinaro et al. [9], [15]. Acyclic activities, such as squatting, and stair-negotiation trials were excluded because they were outside the scope of the present evaluation. Under the participant-level split used for the hip-, knee-, and ankle-moment models (41 training, 5 validation, and 10 test participants), the training partition contained 15.19 h of walking data: 12.06 h of level-ground walking, 1.91 h of ramp ascent, and 1.23 h of ramp descent. The full 56-participant pool contained approximately 20.6 h of retained walking data.

The datasets were aggregated to increase variability across participants, walking speeds, and walking conditions. Level-ground walking provided the largest amount of data and the broadest speed coverage (aggregating both treadmill and overground data), whereas the smaller ramp-ascent and ramp-descent subsets supplied kinematic and joint-moment patterns for evaluation on inclined walking. Before training, all retained

trials were mapped to consistent joint definitions, coordinate conventions, and moment signs. Left- and right-limb data were also transformed to a common limb convention where applicable.

The retained training samples were not uniformly distributed across locomotor conditions: level-ground, ramp-ascent, and ramp-descent walking accounted for 84.3%, 10.7%, and 5.0% of the available windows, respectively. To prevent the larger level-ground subset from dominating model optimization, training windows were selected using balanced sampling across locomotor conditions, ensuring that level-ground walking, ramp ascent, and ramp descent contributed equally to the training loss. Thus, the ramp data received greater sampling weight relative to their raw duration because they contained condition-specific information that was less abundant in the aggregated database. No dataset-level ablation was performed, so the relative usefulness of the three source datasets was not ranked independently.

Sagittal-plane joint angles and net joint moments were extracted from the processed motion-capture and ground-reaction-force data. Joint angular velocities were then computed from the filtered joint-angle trajectories: for planar kinematic chains such as the knee and ankle, each angle time series was fit with a cubic spline and differentiated analytically; for spatial kinematic chains such as the hip, Euler angles were converted to rotation matrices, finite-differenced in time, and converted to angular velocity. Joint moments were normalized by participant body mass (Nm/kg). Lower-body kinematics sampled at 200 Hz in the open-source data were downsampled to 100 Hz to match the control frequency of the evaluated exoskeletons.

Models received kinematic inputs in physical units and directly predicted mass-normalized joint moments, without input or output normalization.

B. Biological Joint Moment Estimator

We used a causal temporal convolutional network (TCN) [34] to estimate biological joint moments from joint kinematics. The network contained five temporal convolutional blocks with 80 hidden channels, a kernel size of five, and dilation factors of 1, 2, 4, 8, and 16. Each block included causal convolution, batch normalization, rectified linear unit activation, and dropout ($p = 0.1$). The input was a causal window of the preceding 100 samples, corresponding to 1.0 s at 100 Hz.

Models were implemented in PyTorch [35] and trained on a workstation equipped with an Intel Core i9-14900KF CPU (24 cores, 32 threads), an NVIDIA GeForce RTX 4090 GPU (24GB VRAM), and 64GB of system memory, running Ubuntu 22.04. Training used Huber loss with a transition parameter of 0.5 Nm/kg. We used AdamW with a weight decay of 1×10^{-4} , a batch size of 512, and gradient clipping at a norm of 1.0. A cosine-annealing scheduler reduced the learning rate from 2×10^{-5} to 1×10^{-6} . Training continued for up to five epochs, with early stopping based on validation mean squared error and a patience of two epochs. Separate models were trained using single-joint inputs and bilateral hip, knee, and

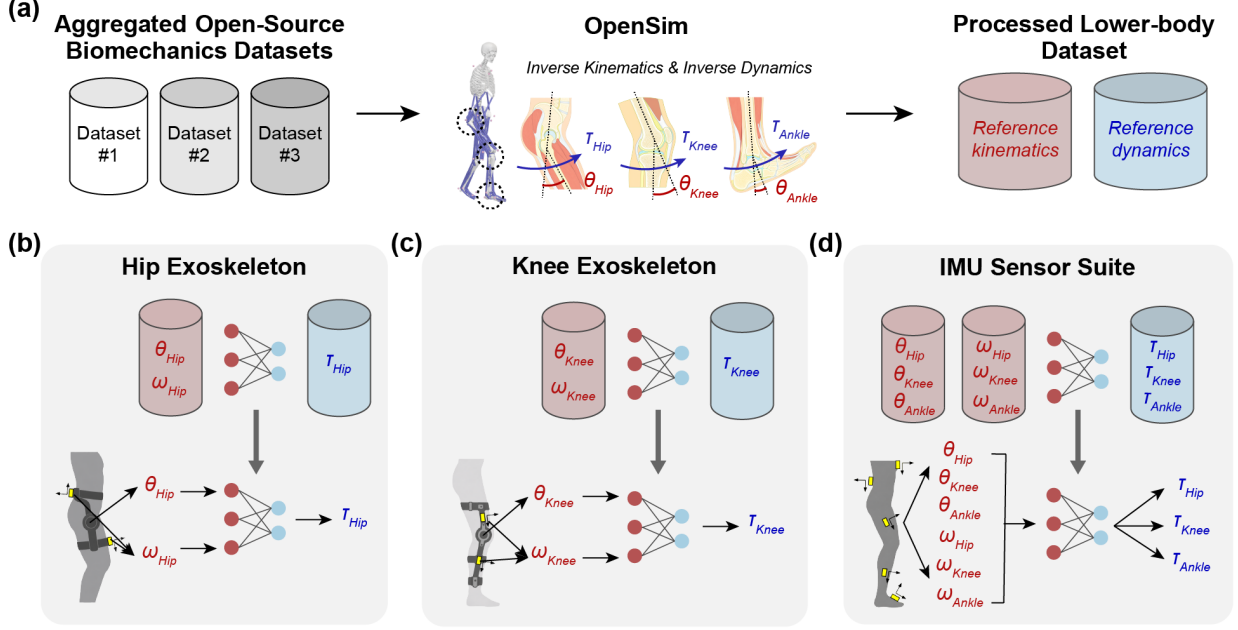


Fig. 1. Overview of the proposed kinematics-based framework for reusable biological joint moment estimation across wearable systems. (a) Open-source biomechanics datasets are aggregated and processed using OpenSim inverse kinematics and inverse dynamics to obtain reference lower-body kinematics and joint moments for model training. The resulting models are then applied to wearable-derived kinematics from (b) a hip exoskeleton, using hip angle and angular velocity (θ_{Hip} , ω_{Hip}) to estimate the hip moment (T_{Hip}); (c) a knee exoskeleton, using knee angle and angular velocity (θ_{Knee} , ω_{Knee}) to estimate the knee moment (T_{Knee}); and (d) an inertial measurement unit (IMU) sensor suite, using multi-joint lower-body angles and angular velocities to simultaneously estimate hip, knee, and ankle moments (T_{Hip} , T_{Knee} , and T_{Ankle}).

ankle inputs. The trained models were converted to TensorRT engines [36] for GPU-accelerated edge inference.

C. Wearable Sensing and Exoskeleton Platforms

1) *Robotic Hip Exoskeleton*: Hip-exoskeleton experiments utilized a bilateral device (4.5 kg) developed for sagittal-plane assistance [37], featuring brushless actuators (AK80-9, CubeMars, China) with integrated motor encoders and an onboard computing system (Jetson Orin Nano, NVIDIA, USA) (Supplementary Table S2, Supplementary Figure S1(a)). Hip angle and angular velocity, derived from the encoders and relative movement of pelvis- and thigh-mounted IMUs, were provided to the shared single-joint hip-moment estimator (Fig. 1(b)). Output torque for each leg was commanded as:

$$\text{clip} \left[\mathcal{F}_{\text{out}} \left(\alpha_{\text{hip}} \cdot m \cdot \hat{M}_{\text{bio}}(t - \Delta t_{\text{hip}}) \right), -\tau_{\text{lim}}, \tau_{\text{lim}} \right], \quad (1)$$

where $\hat{M}_{\text{bio}}$ is the predicted biological hip moment, m is participant body mass, $\alpha_{\text{hip}} = 0.12$ is the torque scaling factor, $\Delta t_{\text{hip}} = 120$ ms is the command delay, $\mathcal{F}_{\text{out}}$ represents filtering operations, and $\tau_{\text{lim}} = 15$ Nm is the torque limit.

2) *Robotic Knee Exoskeleton*: Knee-exoskeleton experiments utilized a unilateral quasi-direct-drive device worn on the right leg. The actuated knee module (1.3 kg) and a waist-mounted electronics module (0.9 kg) communicated at 100 Hz. Knee angle was obtained from the motor encoder, while angular velocity was estimated from thigh- and shank-mounted IMUs (Fig. 1(c), Supplementary Figure S1(b)). The knee controller mirrored the structure of Eq. 1, but utilized a scaling factor of $\alpha_{\text{knee}} = 0.09$ with no additional command delay. Because commanded motor torque varied with

participant mass, the scaling factor was selected to keep peak torque below approximately one-half of the motors' peak instantaneous torque capacity. This range also limited torque-induced motor displacement, while behavior at higher torque levels was beyond the scope of this study. The onboard control architecture for both exoskeletons is outlined in Supplementary Figure S1(c).

3) *IMU Sensor Suite*: A wireless inertial sensing system [38] (MTw Awinda, Movella, Netherlands) provided unassisted, bilateral, multi-joint kinematics (Fig. 1(d)). Sensors were affixed to the posterior pelvis, lateral thighs, lateral shanks, and dorsal feet (Supplementary Figure S1(d)). Accelerometer, gyroscope, and magnetometer measurements were fused using the versatile quaternion-based filter (VQF) [39] to estimate sensor orientations, which were then used to reconstruct joint kinematics with the IMoveLab workflow [40]. Angular velocities were also computed using centered finite differences.

D. Participants

Eight healthy adults (six males, two females; age: 26.3 ± 2.3 years; height: 1.74 ± 0.06 m; body mass: 72.3 ± 13.3 kg) participated in this study (Supplementary Table S1). Participants were excluded if they presented with a current musculoskeletal injury, neurological disorder, cardiovascular condition, or any other impairment that could affect walking or safe completion of the experimental protocol. All participants provided written informed consent prior to participation, and the study protocol (Protocol #STUDY2026_00000119) was approved by the Carnegie Mellon University Institutional

Review Board. Each participant completed the experimental protocols using the hip exoskeleton, knee exoskeleton, and inertial measurement unit (IMU) sensor suite in one experimental day.

E. Experimental Protocol

Experiments were conducted on a force-instrumented split-belt treadmill (Bertec, USA). Participants performed level-ground walking (LG) at 0.8, 1.2, and 1.6 m/s, ramp ascent (RA, $+10^\circ$) at 0.8 m/s, and ramp descent (RD, -10°) at 0.8 m/s. Each trial lasted approximately 80 s, with the first and last 10 s trimmed to isolate steady-state gait. We analyzed 31 hip exoskeleton trials, 16 knee exoskeleton trials, and 39 IMU sensor suite trials, excluding specific 0.8 m/s and 1.6 m/s LG trials due to corrupted reference OpenSim data.

F. Reference Biomechanics and Moment Estimation

Reference kinematics were collected at 100 Hz using a 24-marker optical motion-capture system (Vero, Vicon, UK) with 10 cameras. Participant-specific musculoskeletal models based on the OpenSim Gait 2392 model [41]–[43] were scaled for each session using OpenSim 4.5 [44], directly incorporating the added mass and inertia of the hip (2.0 kg pelvis, 0.5 kg per femur, 1.3 kg torso) or knee (0.9 kg pelvis, 0.91 kg right femur, 0.39 kg right tibia) exoskeletons. Inverse kinematics coordinates were low-pass filtered at 6 Hz before calculating net sagittal-plane joint moments via inverse dynamics. For assisted trials, biological joint moments were estimated by aligning and subtracting the measured exoskeleton motor torque from the OpenSim-derived net joint moment prior to normalization by the participant's body mass.

G. Model Evaluation

Model performance was evaluated using mass-normalized Root Mean Square Error (RMSE) and the coefficient of determination (R^2) between predicted and reference moments.

Unless specified, results are reported as mean $\pm$ standard deviation. The hip-exoskeleton analysis included 31 trials, with one level-ground trial at 1.6 m/s excluded due to failed inverse dynamics stemming from poor marker data. The knee-exoskeleton analysis included all 16 collected trials. The IMU sensor suite analysis included 39 trials, with one level-ground trial at 0.8 m/s excluded for the same reason. Detailed joint moment prediction metrics across all joints, conditions, and platforms are summarized in Table I.

For the IMU sensor suite trials, joint moment estimates from the multi-joint and per-joint models were statistically compared using paired one-sided t -tests across six joint-level metrics (RMSE and R^2 for the hip, knee, and ankle; H_1 : per-joint performance is inferior to multi-joint performance). Holm–Bonferroni correction was applied across the six comparisons with a significance level of $\alpha = 0.05$.

III. RESULTS

A. Hip Exoskeleton Moment Estimation

The kinematics-based estimator accurately tracked biological hip moments across level-ground walking speeds and

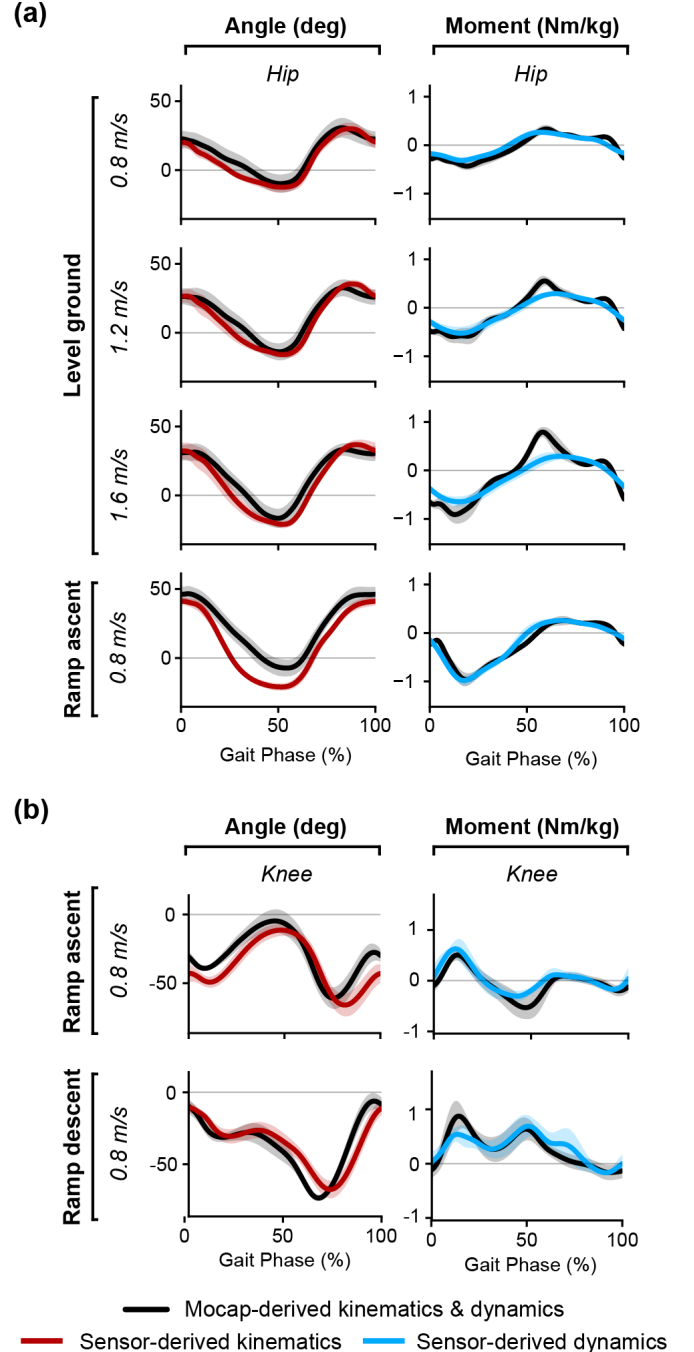


Fig. 2. Wearable-derived kinematics and biological joint moment estimates across the gait cycle for the hip and knee exoskeletons. (a) Hip results during level-ground walking at 0.8, 1.2, and 1.6 m/s and ramp ascent at 0.8 m/s (hip flexion angle is positive). Hip angle was measured using the motor encoder, and hip angular velocity was derived from the pelvis and thigh inertial measurement units (IMUs). (b) Knee results during ramp ascent and descent at 0.8 m/s (knee flexion angle is negative). Knee angle was measured using the motor encoder, and knee angular velocity was derived from the thigh and shank IMUs. Black curves show motion-capture-derived reference kinematics and inverse-dynamics reference biological moments, red curves show wearable-derived joint angles, and blue curves show biological joint moments estimated from the wearable-derived kinematics. Solid curves represent mean trajectories, and shaded regions denote one standard deviation across participants. Gait cycles are segmented by heel-strike gait events.

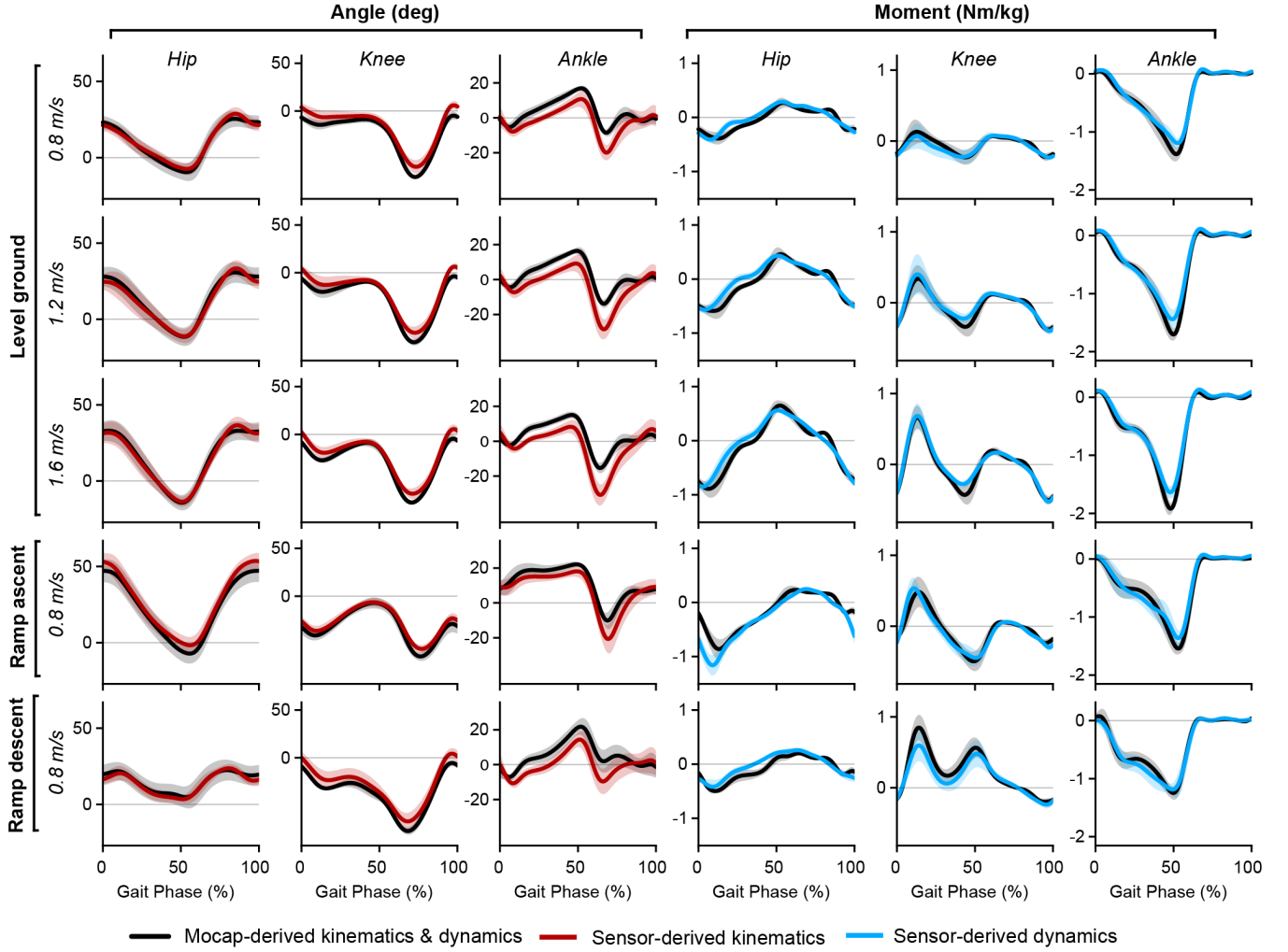


Fig. 3. Lower-body kinematics and biological joint moment estimates across the gait cycle using the inertial measurement unit (IMU) sensor suite. Results are shown for the hip, knee, and ankle during level-ground walking at 0.8, 1.2, and 1.6 m/s and ramp ascent and descent at 0.8 m/s. Black curves show reference kinematics obtained from motion-capture-based inverse kinematics and reference moments obtained from inverse dynamics. Red curves show IMU-derived joint angles, and blue curves show the biological joint moments estimated from the IMU-derived multi-joint kinematics. Solid curves represent mean trajectories, and shaded regions denote one standard deviation across participants.

ramp ascent using motor-encoder hip angles and IMU-derived angular velocities (Fig. 2(a)). Across all four conditions, mean RMSE ranged from 0.12 ± 0.04 to 0.25 ± 0.08 Nm/kg, and mean R^2 ranged from 0.76 ± 0.10 to 0.84 ± 0.09 . During level-ground walking, mean RMSE increased with speed from 0.12 ± 0.04 Nm/kg at 0.8 m/s to 0.25 ± 0.08 Nm/kg at 1.6 m/s. The highest tracking fidelity was observed during ramp ascent ($R^2 = 0.84 \pm 0.09$; Table I).

B. Knee Exoskeleton Moment Estimation

Using knee angle obtained from the motor encoder and knee angular velocity derived from the thigh and shank IMUs, the estimator predicted knee moments during ramp ascent and ramp descent (Fig. 2 (b)). Mean RMSE was higher during ramp descent than during ramp ascent (0.22 ± 0.03 versus 0.17 ± 0.03 Nm/kg), while mean R^2 was lower during ramp descent (0.58 ± 0.14) than during ramp ascent (0.65 ± 0.14 ; Table I).

C. Multi-Joint Moment Estimation Using IMU Sensor Suite Kinematics

Using bilateral hip, knee, and ankle angles and angular velocities derived from the IMU sensor suite, the estimator predicted multi-joint moments across level-ground walking speeds, ramp ascent, and ramp descent. Across the 15 joint-condition combinations, mean RMSE ranged from 0.11 ± 0.02 to 0.21 ± 0.06 Nm/kg, and mean R^2 ranged from 0.60 ± 0.25 to 0.97 ± 0.03 . During level-ground walking, mean R^2 increased with speed at all three joints, rising from 0.80 to 0.92 at the hip, from 0.60 to 0.90 at the knee, and from 0.92 to 0.97 at the ankle between 0.8 and 1.6 m/s.

D. Agreement with Marker-based Motion Capture

Across the bilateral hip, knee, and ankle channels, the IMU-derived joint angles demonstrated high agreement with the corresponding inverse-kinematics trajectories (R^2 of 0.919 ± 0.103) from OpenSim, with an overall RMSE of $7.438 \pm$

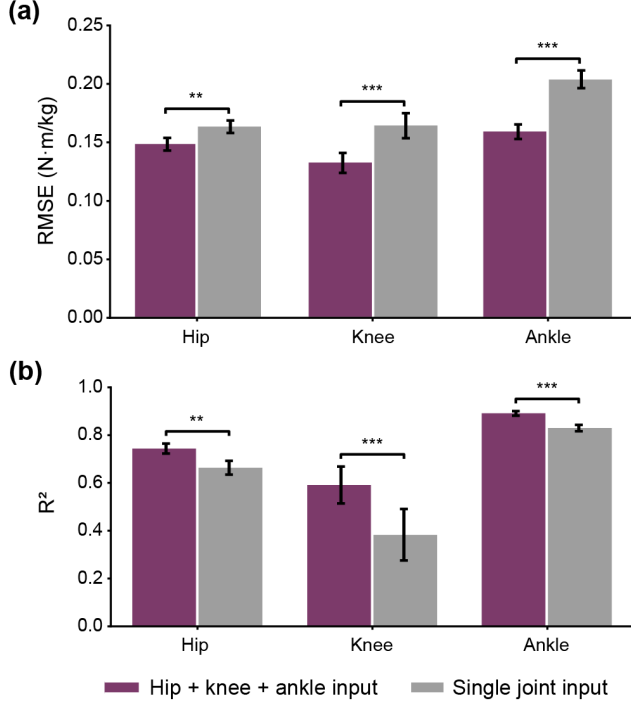


Fig. 4. Comparison of multi-joint and single-joint kinematic inputs for biological joint moment estimation using the IMU sensor suite. (a) Mass-normalized RMSE and (b) R^2 for hip, knee, and ankle moment estimates. The multi-joint model used hip, knee, and ankle angles and angular velocities as inputs, whereas each single-joint model used only the angle and angular velocity of the target joint. Multi-joint inputs resulted in lower RMSE and higher R^2 across all three joints. Asterisks indicate significant paired differences between input configurations (** $p < 0.01$, *** $p < 0.001$).

4.320°. Hip flexion and knee extension angles exhibited the strongest correlation ($R^2 > 0.96$), whereas agreement for ankle dorsiflexion angle was comparatively lower ($R^2 \approx 0.80$), contributing to the observed variability in downstream moment prediction.

IV. DISCUSSION

The central implication of this study is that cross-device transfer of biological joint moment estimators can be addressed as a representation problem. Raw encoder and inertial signals depend on sensor placement, device geometry, attachment, and signal-processing conventions, making models trained directly on those signals difficult to reuse across hardware. Joint angles and angular velocities instead describe a biomechanical state with a consistent physical meaning across platforms. By placing hardware-specific processing upstream of a common kinematic interface, the proposed framework separates the problem of reconstructing human motion from the downstream estimation of biological joint moments. This decomposition allows the moment estimator to remain unchanged as the sensing platform changes.

A. Joint Kinematics as a Common Representation Across Wearable Platforms

Each platform in this study was responsible only for producing the joint angles and angular velocities expected by

TABLE I
JOINT MOMENT PREDICTION PERFORMANCE ACROSS JOINTS AND TRIAL CONDITIONS.

Trial	Joint	Condition	RMSE (Nm/kg)	R^2
Hip exoskeleton	Hip	LG 0.8m/s	0.12 ± 0.04	0.76 ± 0.13
	Hip	LG 1.2m/s	0.16 ± 0.03	0.81 ± 0.08
	Hip	LG 1.6m/s	0.25 ± 0.08	0.76 ± 0.10
	Hip	RA 0.8m/s	0.16 ± 0.03	0.84 ± 0.09
Knee exoskeleton	Knee	RA 0.8m/s	0.17 ± 0.03	0.65 ± 0.14
	Knee	RD 0.8m/s	0.22 ± 0.03	0.58 ± 0.14
IMU sensor suite	Hip	LG 0.8m/s	0.11 ± 0.02	0.80 ± 0.07
	Hip	LG 1.2m/s	0.14 ± 0.02	0.88 ± 0.04
	Hip	LG 1.6m/s	0.16 ± 0.02	0.92 ± 0.02
	Hip	RA 0.8m/s	0.21 ± 0.06	0.83 ± 0.06
	Hip	RD 0.8m/s	0.13 ± 0.03	0.75 ± 0.10
	Knee	LG 0.8m/s	0.13 ± 0.07	0.60 ± 0.25
	Knee	LG 1.2m/s	0.11 ± 0.04	0.83 ± 0.08
	Knee	LG 1.6m/s	0.12 ± 0.04	0.90 ± 0.05
	Knee	RA 0.8m/s	0.14 ± 0.05	0.78 ± 0.09
	Knee	RD 0.8m/s	0.17 ± 0.10	0.82 ± 0.20
	Ankle	LG 0.8m/s	0.16 ± 0.07	0.92 ± 0.05
	Ankle	LG 1.2m/s	0.14 ± 0.05	0.96 ± 0.02
	Ankle	LG 1.6m/s	0.15 ± 0.05	0.97 ± 0.03
	Ankle	RA 0.8m/s	0.19 ± 0.04	0.90 ± 0.04
	Ankle	RD 0.8m/s	0.16 ± 0.03	0.91 ± 0.03

Values are reported as mean ± standard deviation across participants. LG = level ground; RA = ramp ascent; RD = ramp descent. Walking speeds are reported in m/s. RMSE was normalized by participant body mass and is reported in Nm/kg. Exoskeleton results were calculated using the deployed encoder and IMU-derived kinematic inputs. For the IMU sensor suite trials, left- and right-side metrics were averaged within each participant before calculating group-level means and standard deviations. Rows in bold indicate best-performing condition based on mean R^2 .

the downstream estimator. The hip and knee exoskeletons generated these inputs from combinations of motor encoders and segment-mounted IMUs, whereas the IMU sensor suite reconstructed bilateral, multi-joint kinematics through inertial motion capture. Once expressed in joint coordinates, these otherwise different sensor configurations could interface with models trained using open-source biomechanics data.

This modularity distinguishes the proposed formulation from models that estimate joint moment directly from a fixed set of linear accelerations and angular velocity signals from IMUs. A raw-signal model may implicitly learn device-specific properties such as sensor orientation, attachment dynamics, encoder alignment, and interface motion. Such properties can improve performance on the original device but restrict reuse when the sensor configuration changes. The kinematics-based formulation intentionally delegates these properties to the device-specific front end and retains a common biomechanical representation for moment estimation. Transfer therefore depends primarily on whether a new platform can provide sufficiently accurate and consistently defined joint kinematics, rather than whether it reproduces the original training sensors.

From a deployment perspective, a new wearable platform therefore does not need to reproduce the raw sensor configuration represented in the training data; it only needs to provide joint angles and angular velocities in the expected coordinate convention. This enables the downstream estimator to be trained using open-source biomechanics datasets containing joint kinematics and inverse-dynamics moments, even when those datasets do not include signals from the target device.

The hip and knee exoskeleton experiments demonstrate that this common interface can be generated causally from local onboard sensing, whereas the IMU sensor suite results extend the same formulation to bilateral, multi-joint offline estimation.

Condition-averaged performance is important for control because a deployable estimator must remain reliable as locomotor demands change, rather than performing well in only one task. Averaging performance across level ground, ramp ascent, and ramp descent conditions gives each condition equal interpretive weight and provides a compact measure of cross-condition robustness. In particular, condition-averaged R^2 indicates how consistently the estimator preserves the temporal variation of biological joint moments across different walking contexts. Although this measure does not replace evaluation during unconstrained daily activity, it provides a practical proxy for the estimator's ability to support control across the varied locomotor conditions encountered in everyday mobility.

B. Comparison with Existing Estimation and Domain-Adaptation Approaches

Direct comparisons with prior studies should be interpreted cautiously because the evaluated activities, assistance profiles, sensing configurations, participants, and metric aggregation differed. Molinaro et al. [9] reported cyclic-task RMSE values of 0.15 and 0.13 Nm/kg at the hip and knee, respectively, using labeled data collected from the target exoskeleton. In the present study, hip-exoskeleton RMSE ranged from 0.12 to 0.25 Nm/kg. Knee RMSE ranged from 0.17 to 0.22 Nm/kg.

Scherpereel et al. [17] reduced the need for labeled target-device data through semisupervised and unsupervised domain adaptation. Their semisupervised estimator achieved RMSE values of 0.19 Nm/kg at the hip and 0.16 Nm/kg at the knee. The corresponding unsupervised results were 0.21 Nm/kg and 0.18 Nm/kg.

In addition to predictive accuracy, the computational scale of the present models is comparable to prior real-time joint moment estimators. The single-joint models contained 291,521 trainable parameters and approximately 1.12 Mebibytes (MiB) of 32-bit weights. The multi-joint model contained 293,603 parameters and 1.13 MiB of weights, a 0.7% increase because the models shared the same TCN backbone and differed mainly in their input and output layers. For comparison, the TCN architecture reported by Molinaro et al. [9] is estimated to contain approximately 302,000 parameters and 1.15 MiB of 32-bit weights. Scherpereel et al. [17] similarly deployed only the TCN moment estimator at runtime, although its parameter count was not reported.

The primary advantage of the present framework is therefore reduced dependence on target-device data. Molinaro et al. used labeled target-exoskeleton data, while Scherpereel et al. used unlabeled target-device recordings to learn a device-specific translation. In contrast, none of the evaluated hip-exoskeleton, knee-exoskeleton, or IMU sensor suite data were used for training, adaptation, fine-tuning, or checkpoint selection. Transfer was achieved by converting each platform's measurements into the joint-kinematic representation available in open-source biomechanics datasets.

These approaches are complementary. Domain adaptation learns a mapping between source- and target-device signal distributions and may preserve information in the target device's raw signals, but it requires target-domain recordings and additional translation models. The present framework instead defines joint kinematics as the common interface, simplifying cross-device deployment while making performance dependent on the accuracy of each platform's kinematic estimates.

C. Influence of Kinematic Input Representation and Domain Alignment

Evaluating the model with reference motion-capture-derived kinematics yielded closer performance to the training baseline across all platforms. For the hip exoskeleton, RMSE decreased by 10.0% and R^2 increased by 0.034. A larger change was observed for the knee exoskeleton, where RMSE decreased by 27.4% and R^2 increased by 0.152. Similarly, for the IMU sensor suite, reference kinematics reduced RMSE by 18.2% and increased R^2 by 0.059.

These improvements highlight the sensitivity of the estimator to input domain shifts between onboard measurements and the marker-based kinematics used during training. Because the downstream joint-moment estimator was trained exclusively on optical motion-capture kinematics, input signals generated via motion capture closely match the learned feature distribution.

In contrast, onboard measurements introduce domain-specific variations: exoskeleton encoders capture motion between mechanical links rather than anatomical segments, and wearable IMUs are subject to soft-tissue artifact and sensor alignment offsets. Additionally, because both the input kinematics and target joint moments in the training set were derived from optical marker trajectories, the neural network may have learned specific features implicit to marker-based kinematic reconstructions. Consequently, performance differences between onboard and reference inputs reflect both hardware-sensing deviations and the inherent alignment between optical kinematic representations and the target joint moments.

D. Benefits of Multi-Joint Kinematic Context

For the IMU sensor suite trials, joint moment estimates from both a multi-joint and per-joint model were statistically compared using paired one-sided tests across six joint-level metrics (RMSE and R^2 for the hip, knee, and ankle; H_1 : per-joint performance is inferior to multi-joint performance). Because Shapiro-Wilk tests rejected normality for four of the six paired-difference distributions, non-parametric paired one-sided Wilcoxon signed-rank tests with Holm-Bonferroni correction ($\alpha = 0.05$) were used; parametric paired t -tests yielded identical statistical conclusions.

Compared to single-joint models, the multi-joint model reduced mean RMSE by 17.2% and yielded statistically significant improvements in both RMSE and R^2 across all three joints (Fig. 4). These findings demonstrate that neighboring kinematic channels carry complementary information that cannot be captured from single-joint trajectories alone.

Biological joint moments are not determined in isolation; adjacent-joint kinematics provide crucial context regarding whole-limb configuration and gait progression. This intersegmental coupling provides a mechanistic basis for the performance gains observed when combining hip, knee, and ankle kinematics rather than evaluating the target joint in isolation.

E. Study Impact and Limitations

This study demonstrates that biological joint moment estimators trained entirely on open-source biomechanics data can be deployed across multiple wearable platforms without target-device training, adaptation, fine-tuning, or model selection. Hip-estimation performance remained within the range of prior device-specific and domain-adapted approaches, although lower knee R^2 values indicate that eliminating target-domain adaptation does not ensure equivalent performance across all joints and conditions.

Several limitations should be considered. First, both exoskeletons used quasi-direct-drive actuation and were developed within the same laboratory. Despite differences in the actuated joint and sensing configuration, they share aspects of their mechanical design, interfaces, control architecture, and signal processing. Validation on commercial systems and devices developed by independent laboratories is therefore needed to establish broader hardware generalizability.

Second, assistance torques generally peaked between 5 and 7 Nm, below the devices' approximate 18 Nm capacity. Higher assistance may produce greater biomechanical changes, interface deformation, human–robot misalignment, and sensor vibration. The assisted-versus-unassisted comparison was also confounded by differences between the exoskeleton and IMU sensing pipelines and should therefore be interpreted as a cross-platform comparison rather than a controlled effect of assistance.

Finally, evaluation was limited to periodic level-ground, ramp-ascent, and ramp-descent walking. Performance should not be assumed to generalize to stairs, transitions, turning, obstacle negotiation, or nonlocomotor activities. Expanding the training data to include these behaviors and incorporating complementary measurements, such as pressure-insoles, may broaden the supported task range and provide information about contact state and limb loading that is not fully captured by joint kinematics.

V. CONCLUSION

This study demonstrated that joint kinematics can provide a common intermediate representation for biological joint moment estimation across a hip exoskeleton, knee exoskeleton, and IMU sensor suite. The proposed formulation separates hardware-specific kinematic reconstruction from the downstream joint moment estimation, allowing estimators trained exclusively on open-source biomechanics data to be applied without target-device training or fine-tuning. Hip-estimation performance was within the range reported for existing device-specific and domain-adapted estimators, whereas knee performance identified remaining sensitivity to kinematic input quality and locomotor condition. Multi-joint inputs further

showed that broader limb context can improve estimation when additional sensing is available. Together, these findings support a modular alternative to collecting a new labeled biomechanics dataset for every wearable platform, while emphasizing that accurate kinematic reconstruction, validation under assistance, and testing across more diverse devices and movements remain necessary for broader deployment.

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